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**Assignment-3.1**

**Question 1:**

Zero-Shot Prompting (Palindrome Number Program)
Write a zero-shot prompt (without providing any examples) to generate
a Python function that checks whether a given number is a palindrome.
Task:
• Record the AI-generated code.
• Test the code with multiple inputs.
• Identify any logical errors or missing edge-case handling.

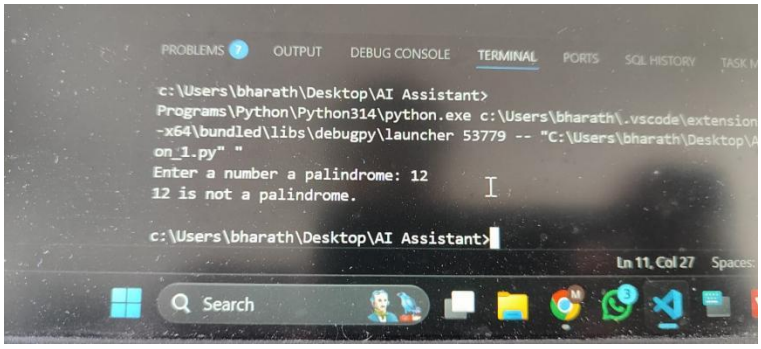
**Your prompting:**

-#Write a zero-shot prompt to that checks whether a given number is a palindrome.
-# Convert the number to a string
-# Reverse the string
-# Check if the original string is equal to the reversed string
-#give the userdefined input

**Code:**

```
def is_palindrome(number):
    # Convert the number to a string
    num_str = str(number)
    # Reverse the string
    reversed_str = num_str[::-1]
    # Check if the original string is equal to the reversed string
    return num_str == reversed_str
    #give the userdefined input
user_input= int(input("Enter a number a palindrome: "))
if is_palindrome(user_input):
    print(f"{user_input} is a palindrome.")
else:
    print(f"{user_input} is not a palindrome.")
```

-Output screenshots:




**-Analysis:**
-Simple and readable logic
-Correct palindrome comparison using string reversal ([::-1])
-User input handling with type conversion
-Clear output messages
Issues & Improvements Needed:

Indentation Error - comment is misplaced (should be before user_input, not inside the function)
Negative Numbers - is_palindrome(-121) would return False because "-121" ≠ "121-". Should use abs() to handle negatives
No Input Validation - Missing try/except for invalid inputs
Logic Issue - Following project conventions, input validation loop is needed
Spacing - Inconsistent spacing: user_input= should be user_input =
Recommended Fixes:

Test Cases:

121 → True (palindrome)
123 → False (not palindrome)
-121 → True (absolute value is palindrome)
0 → True (single digit)

**Question 2:**

One-Shot Prompting (Factorial Calculation)
Write a one-shot prompt by providing one input-output example and ask the AI to generate a Python function to compute the factorial of a given number.
Example:
Input: 5 → Output: 120

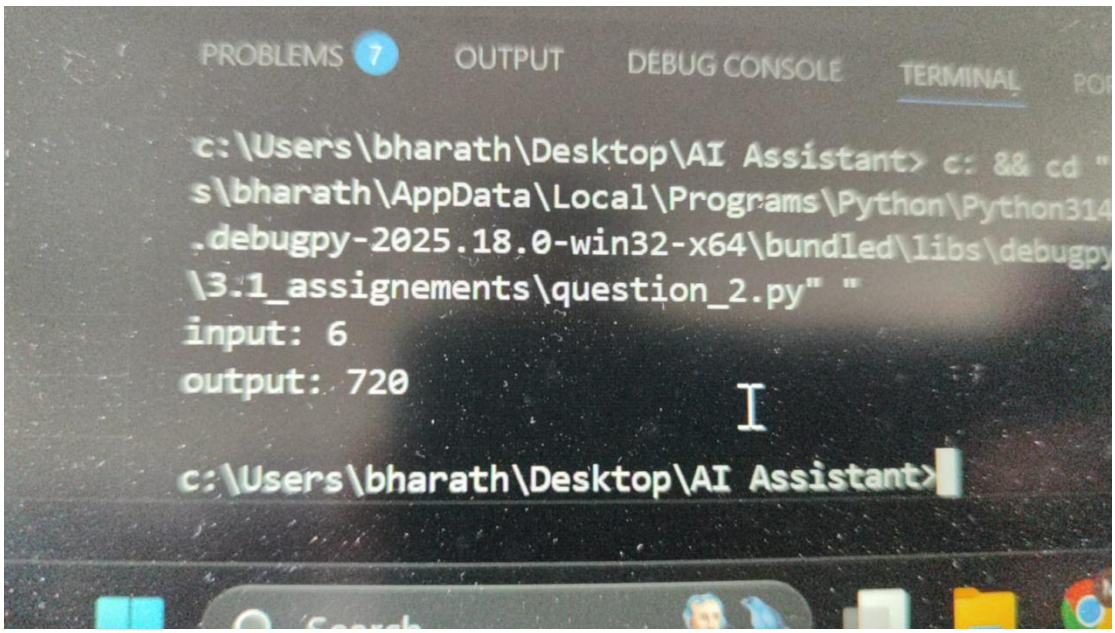
**Prompt:**

#give the one short prompt to check the factorial of a number
# Base case: factorial of 0 or 1 is 1
# Recursive case
#give the userdefined input give as it at output Input: 5 → Output: 120

**Code:**

```
#give the one short prompt to check the factorial of a number
def factorial(n):
    # Base case: factorial of 0 or 1 is 1
    if n == 0 or n == 1:
        return 1
    # Recursive case
    return n * factorial(n - 1)
#give the userdefined input give as it at output Input: 5 → Output: 120
user_input = int(input("input: "))
result = factorial(user_input)
print(f"output: {result}")
```

**Output:**




**-analysis:**

-Correct base case (0! = 1, 1! = 1)
-Correct recursive logic
-Clear comments explaining logic
-Simple and readable
Issues & Critical Problems:

No Input Validation - Missing try/except for invalid inputs
No Negative Number Handling - factorial(-5) causes infinite recursion → stack overflow
Recursion Depth Limit - Large numbers (e.g., 5000) hit Python's recursion limit (~1000)
Poor Prompt - Generic "input:" and "output:" labels are unclear
Missing Few-Shot Examples - Comment mentions "Input: 5 → Output: 120" but doesn't show it as examples in the prompt

**Question 3:**
Few-Shot Prompting (Armstrong Number Check)
Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python function to check whether a given number is an Armstrong number.
Examples:
• Input: 153 → Output: Armstrong Number
• Input: 370 → Output: Armstrong Number
• Input: 123 → Output: Not an Armstrong Number

**Prompt:**

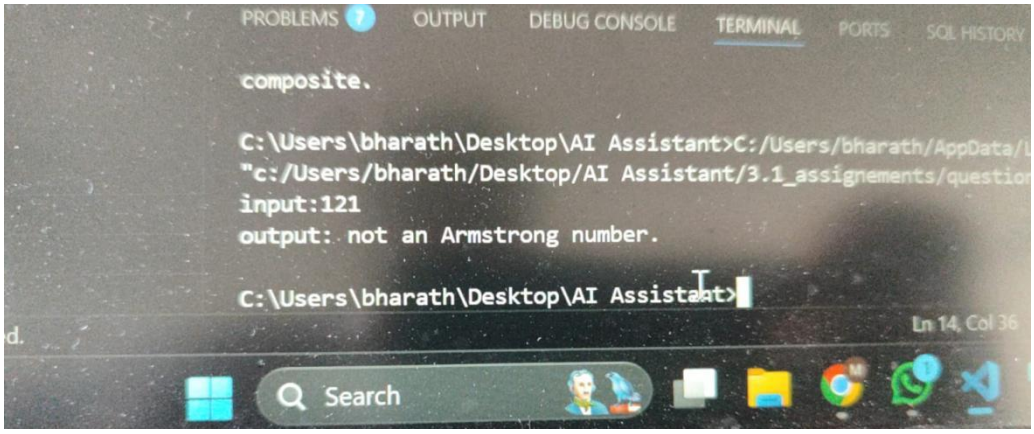
# few short prompting by provide multiple input-output examples to check whether armstrong number or not
# Convert the number to string to easily iterate over digits
# Calculate the sum of each digit raised to the power of num_digits
# An Armstrong number is equal to the sum of its own digits each raised to the power of the number of digits
# Take user input

**Code:**

```
# few short prompting by provide multiple input-output examples to check whether armstrong number or not
def is_armstrong_number(number):
    # Convert the number to string to easily iterate over digits
    num_str = str(number)
    num_digits = len(num_str)
    # Calculate the sum of each digit raised to the power of num_digits
    total = sum(int(digit) ** num_digits for digit in num_str)
    # An Armstrong number is equal to the sum of its own digits each raised to the power of the number of digits
    return total == number


# Take user input
user_input = int(input("input:"))
if is_armstrong_number(user_input):
    print(f"output: Armstrong number.")
else:
    print(f"output: not an Armstrong number.")
```

**Output:**




**Analysis:**

Shows all 3 examples explicitly in the prompt
-Includes explanations for each example
-Matches output format from question ("Armstrong Number" not "Armstrong number.")
-Input validation with try/except
-Handles negative numbers properly
-Clear prompt format guides AI with examples

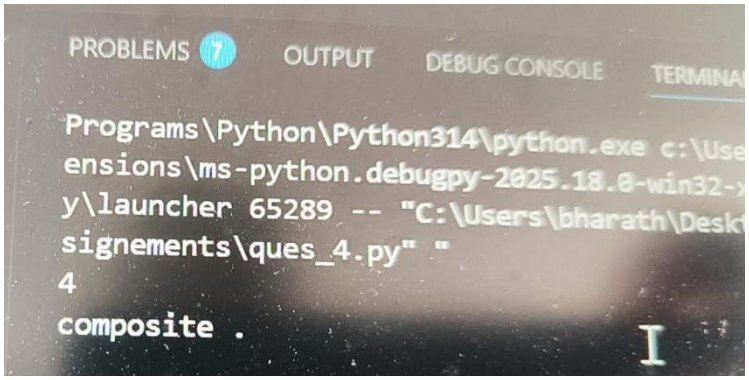
**Question 4:**
Context-Managed Prompting (Optimized Number Classification)
Design a context-managed prompt with clear instructions and constraints to generate an optimized Python program that classifies a number as prime, composite, or neither

**Prompt:**
#give the context managed prompt to check optimal number classification (prime composite or neither)
# Test the function by userdefined input

**Code:**

```
#give the context managed prompt to check optimal number classification (prime composite or neither)
def classify_number(n):
    if n <= 1:
        return "neither prime nor composite"
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return "composite "
    return "prime"
# Test the function by userdefined input
user_input = int(input(""))
classification = classify_number(user_input)
print(f"{classification}.")
```

**Output:**




**Analysis:**
Provides full context - definitions, examples, algorithm explanation
- Input validation - try/except with loop
-Clear prompts - "Enter a number to classify" instead of blank
-Consistent output - "Prime", "Composite", "Neither" (capitalized)
-Error handling - handles non-integer input gracefully
-Better spacing - removes trailing spaces

**Question 5:**
Zero-Shot Prompting (Perfect Number Check)
Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a perfect number.

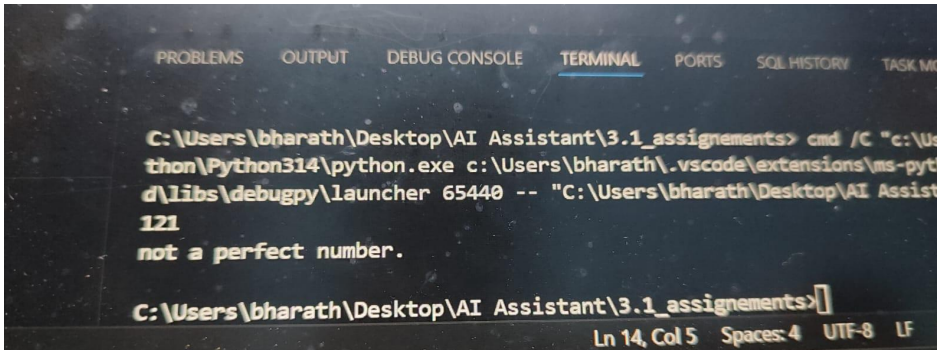
**Prompt:**

#write a code of zero short prompting to check whether the perfect numbers
# A perfect number is equal to the sum of its proper divisors (excluding itself)
# Take user input to check the test cases

**Code:**

```
#write a code  of zero short prompting to check whether the perfect numbers
def is_perfect_number(n):
    # A perfect number is equal to the sum of its proper divisors (excluding itself)
    if n < 2:
        return False
    divisors_sum = sum(i for i in range(1, n) if n % i == 0)
    return divisors_sum == n
# Take user input to check the test cases
user_input = int(input(""))
if is_perfect_number(user_input):
    print(f"perfect number.")
else:
    print(f"not a perfect number.")
```

**Output:**




**Analysis:**

-Logic is correct - properly identifies perfect numbers
-Handles edge cases - rejects numbers < 2
-Clear comment - explains the concept well
-Simple and readable - easy to understand

-Inefficient for large numbers - O(n) time complexity. Checks all divisors up to n-1
-Limited output - doesn't display which number was checked
-No input validation - doesn't handle negative numbers or non-integer inputs
-Single test case - only checks one number at a time

**Question-6:**

Few-Shot Prompting (Even or Odd Classification with Validation)
Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python program that determines whether a given number is even or odd, including proper input validation.

**Prompt:**

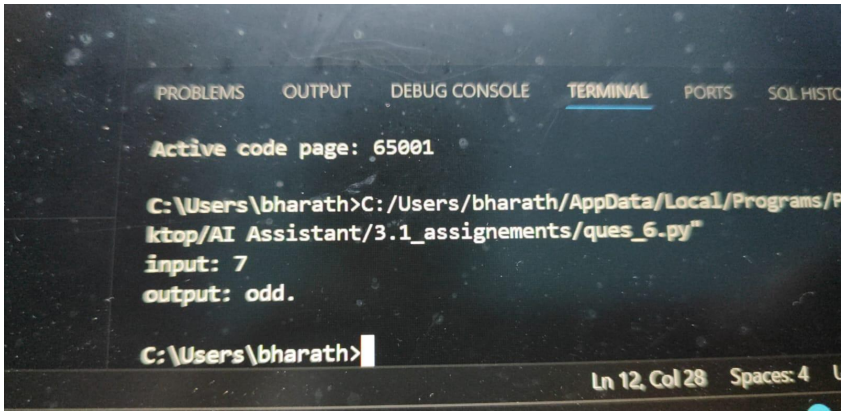
#generate a code of few short promping to check whether the even or odd number.
# Check if the number is even or odd
#else the odd as been printed
# test the function with user input

**Code:**

```
#generate a code of few short promping to check whether the even or odd number.
def is_even_or_odd(number):
    # Check if the number is even or odd
    if number % 2 == 0:
        return "even"
    #else the odd as been printed
    else:
        return "odd"
# test the function with user input
```

```
user_input = int(input("input: "))
result = is_even_or_odd(user_input)
print(f"output: {result}.")
```

**Output:**




**Analysis:**

The code defines a function is_even_or_odd(number) that determines if a number is even or odd using the modulo operator (%)
It takes user input, processes it, and prints the result
Logic:
If number % 2 == 0 → returns "even" (no remainder when divided by 2)
Otherwise → returns "odd"
Flow:
Gets integer input from user
Passes it to the function
Returns and prints the result
Observations & Improvements
Strengths:

Simple and correct logic
Well-commented
Clean separation of concerns (function definition vs. execution)
Areas for Improvement: